Duncan Craine

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EDUCATION

University of Connecticut — Storrs, Connecticut
Ph.D. in Mechanical Engineering — Aug 2026 - Present

- **Research:** Battery Diagnostics · Machine Learning · Large Language Models
- **Current Coursework:** Engineering Analysis · Advanced Solid Mechanics · Graduate Seminar

Connecticut College — New London, Connecticut
B.A. in Computer Science and Statistics & Data Science — Aug 2022 – May 2026

- **Relevant Coursework:** Machine Learning · Time Series Analysis · Generative Data Modeling · Linear Algebra
- **Honors/Awards:** 2x ICSA Academic All-American Sailing Team · 3x All-NESCAC All-Academic Team
  Senior Computer Science Award · Sophomore Computer Science Award

SKILLS

Languages: Python · SQL · R · C++ · Java · JavaScript
ML/AI: PyTorch · TensorFlow · scikit-learn · LLMs · RAG · QLoRA
Data: NumPy · pandas · SciPy · Matplotlib · Plotly
Tools: Git · Docker · Linux · Power BI · Ollama

RESEARCH EXPERIENCE

University of Connecticut – Research Assistant — Storrs, Connecticut
Reliability Engineering and Informatics Laboratory | Computational Engineering and Design Laboratory — Aug 2026 – Present

- Developing **ModularML**, an **open-source**, configuration-driven framework for reproducible ML experimentation, supporting interchangeable model architectures, domain-specific feature engineering, and systematic pipeline evaluation.
- Building a **retrieval-augmented generation (RAG) pipeline** over scientific literature using document parsing, vector embeddings, semantic retrieval, reranking, and LLM-based reasoning to generate physics-informed candidate models.
- Integrating **analytical modeling, machine learning, and optimization** to develop data-efficient approaches for engineering problems where experimental data is limited.
- Conducting **lithium-ion battery aging experiments** using an Arbin battery testing system to collect experimental data for battery health, degradation, and lifetime modeling.

Research Seminar - Connecticut College — New London, Connecticut
LLM-Based NHL Player Summary Generation — Aug 2025 – May 2026

- Fine-tuned **500M–7B parameter language models** using QLoRA to generate structured NHL player summaries from statistical data, adapting models including GPT-2, Mistral, Phi-3, and Qwen.
- Built an **end-to-end ETL and data-generation pipeline** to extract NHL player statistics, generate training summaries, and transform structured data into model-ready prompt/response datasets.
- Designed a **multi-dimensional evaluation framework** using BLEU, chrF++, ROUGE-1/2/L, and PARENT metrics to assess the semantic fidelity, content coverage, and quality of generated player summaries across models.

National Institute for Undersea Vehicle Technology – Summer Undergraduate Research Fellow — Storrs, Connecticut
Reliability Engineering and Informatics Laboratory — May 2025 – Aug 2025

- Conducted 310 **electrochemical impedance spectroscopy tests** on lithium-ion battery cells across controlled states of health, state of charge, and simulated internal short resistance; analyzed experimental data using Python and Power BI.
- Developed and evaluated machine learning models for **internal short-circuit classification** and **resistance estimation**, achieving 98.5% classification accuracy and 8.84 ohm RMSE in resistance regressio**n**.

SELECTED PROJECTS

Formula 1 Unpredictability Consulting — Aug 2025 – Dec 2025

- Data Preparation: Analyzed 74 years of Formula 1 race data spanning 1,125 races, 861 drivers, 212 teams, and 120 variables across 14 relational datasets, integrating and preparing the data for statistical analysis.
- Statistical Analysis & Recommendations: Evaluated factors contributing to Formula 1 race predictability through statistical analysis, providing evidence-based insights into competitive balance and opportunities for improving racing excitement.

LEADERSHIP AND ATHLETICS

Co-president – Sports Analytics Club — Aug 2025 – May 2026

- Led club meetings, recruited members, and organized analytics projects to introduce the skills and knowledge needed for club members to answer their own sports questions.

Varsity Sailing and Club Ice Hockey — Aug 2022 – May 2026